

GROUP 26

SRS DOCUMENT



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Note: this document was originally produced by an eight-member group (Group 26, module 4CPS212). Group membership has since changed; it is now maintained solely by the author listed above, credited as sole author of this revision. This revision corrects the requirements below to match the implemented application (see the revision notes throughout) and removes requirements that described features that do not exist.

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Use Case: Search

Definitions

- Input – a word, phrase or sentence.
- Researcher of Linguistics – the primary user of the corpus, who wishes to understand the workings of the isiZulu language.
- Filters – the Search in / Part of speech / Exact match controls that narrow a search.
- Response – the information the corpus returns for a search, per the chosen filters.
- Suggestion – a similarly-spelled word offered when a search finds nothing.

Purpose

Allow the user to search content within the corpus and receive a response.

Actors

- Researcher of Linguistics

Revision note: The previous version of this document restricted input to “strictly English” in one place and “English and isiZulu” in another — a direct contradiction. The corpus actually holds isiZulu, English, Swati and Sotho fields for every word, and a search matches all four. The requirement below reflects that.

Use Case

- The user enters an input, e.g. “ubuntu”, in the search panel.
- The input may be a single word, a phrase, or a sentence.
- The user's input may match isiZulu, English, Swati or Sotho text in the corpus.
- The user may choose filters (see Use Case: Options).
- The search panel returns a response from the corpus according to the input and the chosen filters.

Alternate Scenario

- The user may enter a misspelled word, phrase or sentence.
- If the search finds no results, the corpus offers up to 5 similarly-spelled suggestions, drawn from the words and translations already in the corpus.
- If none of the suggestions match what the user intended, the response is simply “No results found” — there is no further “linguistic error” classification beyond that.

Trigger

- The user opens the Search page.

Preconditions

- The user must be logged in (every page in the corpus requires an account).
- The search text must not be empty and must contain only letters, numbers, spaces and common punctuation; anything else is rejected before it reaches the database.

Basic Path

- Enter input into the search panel.
- The system applies the chosen filters (Search in / Part of speech / Exact match).
- The response is returned according to those filters.

Alternate Path

- A result's “View Details” link opens the Word Detail page, which shows word usage, phrases and synonyms for that entry (see Use Case: Options).
- Every search is recorded in Search History (see Use Case: History Feature).
- Any two or more words returned by the same search are recorded as a word pair, shown on the Trends page.

Input Requirements

- Input may match isiZulu, English, Swati or Sotho text.
- Input must be 1–100 characters, using only letters, numbers, spaces and common punctuation (' ' - . , ? !).

Error Handling

- Suggestions — for a search with no results, up to 5 similarly-spelled words or translations are suggested.
- Validation error — input containing disallowed characters (for example, HTML tags) is rejected with “Please enter a valid word.” before any database search runs.

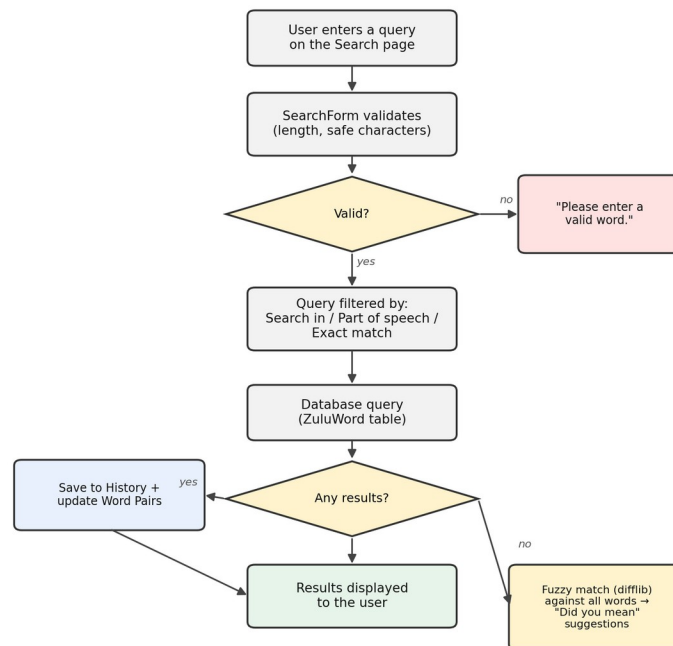
Revision note: The previous version of this section also claimed a fuzzy auto-correcting search (silently correcting a misspelling before searching) and a daily limit on the number of searches. Neither is implemented: a misspelled search returns “did you mean” suggestions to choose from rather than auto-correcting, and there is no daily search limit. Both claims have been removed rather than left in place.

Post Conditions

- The user is shown a response according to the chosen filters, or a suggestion list / validation message if nothing matched.

Figure 1: Search — process flow (as implemented)

Figure 2: Search feature — process flow (as implemented)



Use Case: Favourites Feature

Trigger

The user clicks the star (☆) next to a word in the corpus.

Precondition

The user is logged in and a page showing one or more words with a favourite star is displayed.

Basic Path

- The user finds a word they want to save.
- The user clicks the star next to it.
- The system saves the word to the user's favourites without reloading the page, and the star fills in (★) immediately.
- The user opens the Favourites page from the navigation menu.
- The system displays all saved favourite words, each with its translation, part of speech, pronunciation and cultural context.
- The user may search within their favourites by word or translation.

Alternative Paths

- If the user's session has expired, clicking the star sends them to the login page instead of saving it.
- If the favourites search box contains disallowed characters, the system prompts “Please enter a valid search term.”

Revision note: The previous version of this section described a confirmation dialogue, a success message banner, and a database-connection error message shown to the user. The actual toggle is a single click with no confirmation step, and the endpoint has no user-visible message for a failed save (a network problem simply leaves the star unchanged) — both details have been corrected below.

Postcondition

The word is saved to the user's favourites and can be found, searched, and removed (by clicking the star again) from the Favourites page.

Other

- Favourites are private to the logged-in user.
- Word data (usage, phrases, synonyms, translations) is read live from the corpus database, not cached.

Use Case: Trends Feature

Definition

The Trends feature summarises how the corpus is being used: which words are searched most, and which words tend to appear together in the same search.

Purpose

- Show corpus-wide totals: total words, total phrases, total registered users, total searches.
- Show the most-searched words, as a table and a bar chart.
- Show common word pairs — words that have appeared together in the same search results — with how often each pair has occurred.

Trigger

The user opens the Trends page.

Actors

Linguistic Researchers

Revision note: The previous version of this section described comparing trends “across different corpora, genres or time periods” and analysing change “over time.” Neither is implemented: Trends shows current totals and a fixed top-10 ranking, not a time series, and there is only one corpus, so cross-corpus comparison does not apply. It also stated Total Words and Total Phrases as separate figures without noting that, in the previous implementation, they were always identical (every word was counted as a phrase). Total Phrases is now genuinely a subset: entries whose part of speech is phrase, proverb, idiom or expression, or whose entry contains more than one word.

Preconditions

- The user must be logged in.

Postcondition

The user sees current corpus statistics, the top 10 most-searched words, and the top 10 most frequent word pairs.

Restrictions

- A word's search count only reflects searches performed after this feature was added; it is not backfilled for older data.
- The word-pair count only considers the first 10 results of any one search.

Use Case: Download Feature

Main Function

The download feature lets a logged-in user download the whole corpus as a JSON file, generated from the live database at the moment of download, for offline use or sharing.

Trigger

The user clicks “Download Database (JSON)” on the Search page.

Preconditions

The user must be logged in.

Basic Path

- The user clicks the download button.
- The system builds a JSON file from the current database contents (every word, its translations, examples, and any synonyms and word pairs).
- The browser downloads the file (isiZulu_corpus.json).

Revision note: The previous version of this section described downloading a single selected word or phrase, and linked in the implementation to a static file that was out of date (95 of the corpus's 139 words) and, contrary to this requirement, did not require login at all since static files bypass Django's view-level login check. The feature now always reflects the live database and is enforced server-side, not just described as a requirement.

Alternative Path

If the user is not logged in, the system redirects them to the login page instead of starting the download.

Postcondition

The file is saved to the user's device and can be shared with others or used offline, without needing to log in to read the file itself.

Use Case: Options

Trigger

The user searches for a word in the corpus.

Precondition

The Search page is displayed with a search bar and its filter controls.

Basic Path

- The user enters a word, phrase or sentence in the search bar.
- The user optionally sets Search in (word/translation, phrases & examples, or cultural context), Part of speech, and/or Exact match.
- The user submits the search; the system applies the chosen filters and returns matching entries.
- Each result can be opened (“View Details”) to see its word usage, phrases containing it, its synonyms, and other entries of the same part of speech.

Revision note: The previous version of this section described four independent tick-box options (Word Usage, Translation, Phrases, Synonyms) each producing its own separate result set and its own “no data” message, plus a details view built around those four. The implementation instead uses two search-narrowing filters (Search in, Part of speech) on the Search page itself, and a single Word Detail page (opened per result) that always shows usage, phrases and synonyms together for that word. The requirement below matches that.

Word Detail Page

- Word Usage: other corpus entries whose example sentence contains this word, if any.
- Phrases: proverbs, idioms, expressions or multi-word entries that contain this word, if any.
- Synonyms: entries explicitly linked as synonyms of this word, if any.
- Other entries of the same part of speech, as a browsing aid.

Alternative Paths

- If a section has nothing to show (e.g. no phrases contain the word), the page says so plainly rather than leaving it blank.
- If the search term is invalid (empty, or disallowed characters), the system prompts “Please enter a valid word.”

Postcondition

The user has explored the searched word through its filtered results and, optionally, its detail page.

Use Case: History Feature

Main Function

The History feature keeps track of the words and phrases a user has searched for, so they can return to a past search without retyping it.

Trigger

The user performs a search (recording it), or opens the History page (viewing past searches).

Use Case

- The user searches for a word or phrase.
- The system records the search.
- The user opens the History page.
- The system shows the user's 5 most recent distinct searches, newest first.
- The user selects "Search Again" on a past search to repeat it.

Revision note: The previous version of this section did not specify an order for the history list; the implementation orders it newest-first, capped at 5 distinct queries. It also described repeating a search as moving an existing row to the top; in the implementation, a repeated search is recorded as a new event and older repeats of the same query are then hidden (deduplicated) rather than physically reordered, so the most recent occurrence is what is shown, with the same visible effect.

Restrictions

- History cannot be downloaded; it is only viewable inside the corpus.
- Each user sees only their own history.

Alternative Scenarios

Empty History

If the user has not searched anything yet, the History page shows "No searches found."

Repeated Search

Searching the same term again keeps only its most recent occurrence in the visible list of 5.

Deleting a Search

The user can remove a search from their history; this also removes any earlier occurrences of the same query, and it disappears immediately.

Clearing All History

The user can clear their entire history in one action.

Postconditions

- The user sees their recent searches, newest first.
- A selected past search is repeated and its results shown again.
- Removed searches no longer appear.
- Each user's history remains private to them.

Exception Paths

- Anonymous access: redirected to log in.
- No searches yet: “No searches found.” is shown.

Use Case: Help Feature

Purpose

The Help feature is a guide to using the corpus: clear, immediate instructions for each feature, available from the Help link at any time.

Definitions

- Corpus – the isiZulu Cultural Corpus web application.
- Help Interface – the page where users read this guidance.

Description

The Help page lists every feature in the corpus with a short description of how it works: Search (including its filters and “did you mean” suggestions), Text-to-Speech, Favourites, Trends, Search History, Downloading the Corpus, and Profile.

Revision note: The previous version of this section referred to a search bar “at the top of any page”; the corpus has a Search page reached from the navigation menu, not a persistent search bar on every page. It also described four separate search options (Word Usage / Translation / Phrases & Synonyms as tick-boxes) which have since been corrected to match the real Search in / Part of speech / Exact match filters (see Use Case: Options).

Functional Requirements

- The system shall provide a Help link in the main navigation.
- The Help page shall list every feature in the corpus with a short description.
- The Help page shall be reachable without needing to know where a feature lives first.

Trigger

The user clicks Help in the navigation menu.

Precondition

The user is logged in.

Basic Path

- The user clicks Help.
- The system opens the Help page.
- The Help page displays all features with a short description of each.

Use Case: Profile

Purpose

Lets a user see their own account information and a summary of their activity in the corpus.

What the Profile Page Shows

- Account information: username, email (if provided at registration), and join date.
- Activity summary: number of favourite words, number of searches performed.
- Recent activity: the user's last few searches.
- A logout button.

Revision note: The previous version of this section listed extensive account rules — username/password complexity, date of birth, profile pictures, phone verification, subscription tiers, geographic and device restrictions, and so on. Registration in the implemented system only collects a username and password (Django's built-in user registration form); none of the fields or restrictions above exist, and profile editing is not implemented — the Profile page is read-only. Those requirements have been removed rather than left as unimplemented claims. Anyone designing a future “edit profile” feature should treat the list of possible restrictions as a design reference, not as documentation of current behaviour.

Restrictions (as implemented)

- Username: assigned at registration (Django's default username rules apply); not editable from the Profile page.
- Password: set at registration (Django's default password validators apply: minimum length, not entirely numeric, not too similar to the username).

Trigger

The user clicks Profile in the navigation menu.

Precondition

The user is logged in.

Postcondition

The user sees their account information and activity summary.